

Reza Goharimehr



✉ rgohari1@binghamton.edu

Department of Mechanical Engineering
State University of New York - Binghamton
Binghamton, NY, USA

Education

PhD student in Mechanical Engineering

Binghamton University, State University of New York

August 2022 to Now

Master of Science in Aerospace Engineering - Propulsion

Shahid Beheshti University (SBU), Tehran, Iran

(GPA:16.12/20 The highest rank)

Relevant Modules:

- 1st Class Degree with Advance Numerical Computations Methods (19.45/20)
- 2st Class Degree with CFD 1 (16/20)
- 1st Class Degree with multiphase flow (18.3/20)

Dissertation: **Numerical Simulation of multicomponent multiphase flow transport through a porous media by using Lattice Boltzmann Method**

Sep 2016 to Sep 2019

Bachelor of Science in Mechanical Engineering

Islamic Azad University (IAUN), Najafabad, Isfahan, Iran

(GPA:15.59/20)

Final project: **Simulation of fluid flow around a circular obstacle at various Reynolds numbers via Ansys CFX numerical software**

Sep 2011 to Nov 2015

Work & Research

Teacher Assistant

Thomas J. Watson College of Engineering and Applied Science
Binghamton University, State University of New York

August 2022 to Now

Research Assistant

Multiphase Flow Laboratory, Shahid Beheshti University, Tehran, Iran

Under the supervision of **Dr. Ezzatneshan**

Responsibilities:

- Mentorship undergraduate students in the research process
- Working with students to tackle their problems and address them

June 2018 to Aug 2022

Teacher Assistant

Under the supervision of **Dr. Ezzatneshan**, New Technologies Engineering Department, Shahid Beheshti University, Tehran, Iran

Taught Advance Numerical Computations Methods, a postgraduate course averaging 25 students per semester

Responsibilities:

- Designed lesson plans and class projects
- Deliver a range of teaching and assessment activities including tutorials
- Involved in the development of new materials
- Conducted **FORTRAN**, **Tecplot** and **MATLAB** course for averaging 20 postgraduate students covering basic to intermediate principles

Sep 2017 to June 2018

Publications

Publications

E Ezzatneshan and **R Goharimehr**. " Study of fingering dynamics of two immiscible fluids in a homogeneous porous medium with considering wettability effects using a pore-scale multicomponent lattice Boltzmann model", *European Journal of Computational Mechanics* 2021

E Ezzatneshan and **R Goharimehr**. "A Pseudopotential Lattice Boltzmann Method for Simulation of Two-phase Flow Transport in Porous medium at High-Density and High-Viscosity Ratios", *Geofluids* 2021

E Ezzatneshan and **R Goharimehr**. " Study of spontaneous mobility and imbibition of a liquid droplet in contact with fibrous porous media with considering wettability effects ", *Physics of fluids* 2020

E Ezzatneshan and **R Goharimehr**. " Effect of Multiphase Lattice Boltzmann Collision Models with Single and Multi-Relaxation Times for Simulation of Liquid-Vapor Two-Phase Flows Using Two Different Forcing Schemes" *Mechanical Engineering of Sharif University*. (in Persian) 2019

Conference posters

E Ezzatneshan and **R Goharimehr**., "Study of droplets collision by implementation of multiphase MRT lattice Boltzmann method incorporated with the exact difference method forcing scheme" *Feb. 17-18-2019*
(In Persian)

E Ezzatneshan and **R Goharimehr**., "High viscosity ratio multiphase flow simulation in a porous medium by the Boltzmann method", *March. 26-2020*,
(In Persian)

Skills

Software:

- Ansys Workbench (Expert)
- Fluent (Expert)
- CFX (Expert)
- Tecplot
- Paraview
- nTops
- .

Programing:

- FORTRAN (Expert)
- MATLAB
- C++
- Python

Field of Research & Interests

Multiphase and multicomponent modeling of fluids transport and phenomena

- Fluid flow in porous media and membranes
- Multiphase Heat Transfer
- Cold Plates
- Capillary effect and wettability

Language

English: IELTS: 7.5

Speaking (7)-listening (7.5)-Reading (9)-Writing (7)

Persian: Native

Hobbies & INTERESTS

Reading

Music

Sports
